2.6 Proving Geometric Relationships

Name _____ Period ____ Date _____

Geometry 2.6 · Student sheet

I can…

- use the Right Angles Congruence Theorem.
- use the Congruent Supplements and Congruent Complements Theorems.
- prove and use the Vertical Angles Congruence Theorem.

Vocabulary · fill in the missing word

Flowchart proof — a proof that uses boxes and arrows to show the flow of a logical _____

Paragraph proof — a proof written as _____ in a paragraph

Part 1 · Discovery: supplements of the same angle

Draw two crossing lines. Number the angles 1, 2, 3, 4 going around. Measure ∠1, ∠2, and ∠3.

m∠1	m∠2	m∠3	180° − m∠2

Why must ∠1 and ∠3 be congruent?

Part 2 · Notes and examples (fill in each note, then work the example)

Right Angles Congruence Theorem: all right angles are _____.

Example 1

∠ABC and ∠DEF are right angles. What can you conclude, and why?

Congruent Supplements Theorem: if two angles are supplementary to the same angle (or to congruent angles), then they are _____. The Congruent Complements Theorem works the same way.

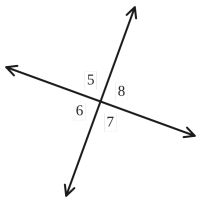
Example 2

∠1 and ∠2 are supplementary, and ∠3 and ∠2 are supplementary. m∠1 = 47°. Find m∠3 and explain in a paragraph proof.

Vertical Angles Congruence Theorem is proved with the _____ Postulate and the Congruent Supplements Theorem.

Example 3

Given: ∠5 and ∠7 are vertical angles. Prove: ∠5 ≅ ∠7.



Example 4

Two lines intersect. Two vertical angles measure $(5x + 10)°$ and $(7x - 18)°$. Find x and all four angle measures.

Part 3 · Partner practice

Partner 1

$\angle 4$ and $\angle 5$ are complementary, and $\angle 6$ and $\angle 5$ are complementary. $m\angle 4 = 28°$. Find $m\angle 6$ and name the theorem.

Partner 2

Two vertical angles measure $(3x + 20)°$ and $(5x - 10)°$. Find x, the angle measure, and the measure of an angle that forms a linear pair with it.